Phantom Experiment Plan

2026-05-09 final - synthesizing rigorous verification + corrected literature

1. Verified by today's rigorous testing

Capabilities verified

Capability	Status	Test
Layer A backward	0.94 ratio at single substep	Phase 2B+2C
Stiffness 1D Adam (5 INITs)	<1% recovery in 1.5s	material_only test
knife_pos vec3 multi-restart	0.04% with median selection	pure_adam test
Material-only single Adam	Robust convergence, no Optuna	verified

Caveats verified

Issue	Caveat	Mitigation
Damping/friction grad	Too weak in single-substep	Use literature (Mackle 2020)
Multi-substep tape	Long-horizon bias	Avoid; short windows only
4D joint local minima	Stuck without restart	Material-only decouple
Cutting saturation	Single-clean insufficient	Use REAL phantom cuts

Literature corrected understanding (verified-read)

Paper	What it actually says	How to cite
Le Lidec 2021	Param obs + cube collision (RIGID body)	Principle yes; inclusion = OUR extrap
Antonova 2022 BO-Leap	BO + CMA-ES + grad for rugged landscape	Cite for non-convex justification only
SliceIt 2024	Optuna+Adam two-stage cutting	Reuse only IF needed (likely not)
DiSECt 2021	Pure Adam through diff cutting	Cite for pure-Adam standard
70% sys-ID papers	Pure Adam single-stage	Field default - our path

2. 18-Week Plan to ICRA 2027 (deadline Sep 15 2026)

Week 1 (this week): Hardware order + paper outline

- Order: Silicone Smooth-On (~$50) + load cell + ADC ($150-200)
- SKIP ABS inclusion initially (our extrapolation, fail-fast first)
- Paper sec3 background outline (DiSECt/SliceIt/Antonova/PAC-NeRF)
- Paper sec4 method outline (Phase 2B+2C verified results)

Week 2-3: Phantom prep + Stage 2.1 POC

- Mold 3-5 specimens with varied mix ratios (different stiffness)
- (If accessible) MRE measurement for GT stiffness reference
- Real protocol: robot pushes knife shallow (no deep cut)
- Record force(t) at 500 Hz via load cell
- Run verified single-substep stiffness Adam calibrator

Pass criterion: stiffness recovery <10% vs MRE measurement

Week 4-5: Stage 2.2 Cutting Calibration

- Real cuts: robot cuts 5-10cm into phantom (deep, real damage)
- Post-cut: 3D scan or photo cavity geometry
- Calibrate alpha_min + sharpness via Stage B framework

Note: today's saturation issue was test artifact (sim from clean state, shallow knife) - real cuts produce significant damage, alpha not saturated

Pass criterion: cavity IoU > 60%, sharpness identifiable to <15%

Week 6: Stage 2.3 Validation Gate (fail-fast)

- 70/30 train/test split
- 4 metrics: IoU, force-trace MSE, sharpness, mass conservation

IF FAIL -> reconsider: ABS inclusion (our extension) / different protocol / accept limitations

Week 7-10: Stage 3 Open-loop Replay (ICRA bar)

- Optionally retrain v4 PPO under calibrated physics (6 GPU hr)
- OR skip if v3 eval shows <5% degradation in sigmoid sim
- Compare sim trajectory vs real (force trace + cavity)

Pass: ~15% match to real (Liu 2022 RAL bar)

Week 11-13: Stage 4 Closed-loop POC (stretch)

- Real robot debulks real phantom using policy
- Goal: 9/9 non-catastrophic outcomes

Week 14-18: Paper writing + submit

- Week 14: sec4 method + sec5 experiments draft
- Week 15: sec1 intro + sec2 related + sec3 background
- Week 16: figures + integration review
- Week 17: external review + revise
- Week 18: submit ICRA Sep 15

3. Strategic Decisions Committed (2026-05-09)

Decision 1: Inclusion or no inclusion?

No inclusion initially. Fail-fast at Stage 2.3 gate.

Rationale: ABS inclusion is OUR extrapolation (not Le Lidec verified); cheaper to verify baseline first; SliceIt also uses homogeneous phantom.

Decision 2: v3 PPO retrain or skip?

v3 eval first -> decide.

- Run v3 checkpoint in current sigmoid sim, compare metrics
- If <5% degradation: skip retrain (saves 6 GPU hr)
- If >=5% degradation: retrain v4

Decision 3: Damping/friction strategy

Literature values (Mackle 2020 PVA cryogel reference).

- Single-substep gradient too weak; multi-substep biased; literature values stable
- Cite Mackle 2020 in paper sec4

4. Risks + Mitigations

Risk	Likelihood	Mitigation
Stage 2.1 fails on real noisy data	Medium	Multiple specimens for variance
Stage 2.2 cavity scan accuracy	Medium	Depth camera + multi-angle
MRE access unavailable	Low	Fall back to spec stiffness
Phantom too soft/stiff	Medium	3-5 specimens cover range
Robot encoder noise	Low	Average multiple reps
ICRA timeline pressure	Medium	Stage 4 stretch; Stage 1+2+3 is bar

5. Acceptance Bar for ICRA Submission

Must-have (Stage 1+2+3)

- Verified differentiable simulator (DONE today)
- Sys-ID framework (today's plan)
- Calibration on >=3 phantom specimens
- Open-loop replay match within 15% on real cuts

Stretch (Stage 4)

- Closed-loop policy on real robot
- 9/9 non-catastrophic safety bar